SELF ASSESSMENT TERM 1 MODEL PAPER - 2026 - 2027 | SET - 1

SOLUTIONS (English Medium)

Class : 7 | Subject : Maths | No. of Questions : 33 | Total Marks : 80 | Duration : 3 Hrs

Section A : Multiple Choice Questions - Each question has four options out of which ONLY ONE option is correct. 20 x 1 = 20

1. The number name of 5,04,085 is **[B]**

A) five lakh forty thousand eighty five | B) five lakh four thousand eighty five | C) fifty lakh four thousand eighty five | D) five lakh four hundred eighty five

[Chapter 1 : Large Numbers Around Us]

Solution :

In the Indian system 5,04,085 is read from the left: 5 lakh, 04 thousand, 085.
5 lakh = 5,00,000 ; 04 thousand = 4,000 ; 085 = eighty five.
So the number name is five lakh four thousand eighty five.
Answer : B) five lakh four thousand eighty five

2. 1 crore = _____ lakhs **[B]**

A) 10 | B) 100 | C) 1000 | D) 10,000

[Chapter 1 : Large Numbers Around Us]

Solution :

1 crore = 1,00,00,000 and 1 lakh = 1,00,000.
1,00,00,000 ÷ 1,00,000 = 100.
So 1 crore = 100 lakhs.
Answer : B) 100

3. In the international (American) system, 1,00,00,000 is written as **[A]**

A) 10,000,000 | B) 100,000,000 | C) 1,000,000 | D) 1,000,000,000

[Chapter 1 : Large Numbers Around Us]

Solution :

1,00,00,000 (one crore) has 8 digits and equals ten million.
In the international system commas are placed after every three digits from the right: 10,000,000.
Answer : A) 10,000,000

4. The value of $18 + 4 \times 5$ is **[B]**

A) 110 | B) 38 | C) 90 | D) 27

[Chapter 2 : Arithmetic Expressions]

Solution :

Multiplication is done before addition.
$4 \times 5 = 20$.
$18 + 20 = 38$.
Answer : B) 38

5. The expression $14 - (12 + 10)$ is equal to **[C]**

A) $14 + 12 + 10$ | B) $14 - 12 + 10$ | C) $14 - 12 - 10$ | D) $14 + 12 - 10$

[Chapter 2 : Arithmetic Expressions]

Solution :

When a bracket has a minus sign before it, every term inside the bracket changes its sign when the bracket is removed.
$14 - (12 + 10) = 14 - 12 - 10$.
Check: $14 - 22 = -8$ and $14 - 12 - 10 = -8$.
Answer : C) $14 - 12 - 10$

6. The value of $7 \times 45 + 3 \times 45$ is **[B]**

A) 315 | B) 450 | C) 135 | D) 495

[Chapter 2 : Arithmetic Expressions]

Solution :

Use the distributive property: $7 \times 45 + 3 \times 45 = (7 + 3) \times 45$.
$(7 + 3) \times 45 = 10 \times 45 = 450$.
Answer : B) 450

7. The place value of 5 in the decimal number 7.05 is **[C]**

A) 5 ones B) 5 tenths C) 5 hundredths D) 5 thousandths

[Chapter 3 : A Peek Beyond the Point]

Solution :

In 7.05 the digits after the decimal point are 0 (tenths place) and 5 (hundredths place).
So the digit 5 is in the hundredths place and its place value is 5 hundredths (0.05).
Answer : C) 5 hundredths

8. 5.6 cm = _____ mm **[B]**

A) 5.6 B) 56 C) 560 D) 0.56

[Chapter 3 : A Peek Beyond the Point]

Solution :

1 cm = 10 mm.
5.6 cm = 5.6 × 10 mm = 56 mm.
Answer : B) 56

9. Which of the following is the greatest? **[C]**

A) 4.5 B) 4.05 C) 4.505 D) 4.055

[Chapter 3 : A Peek Beyond the Point]

Solution :

Write all numbers with three decimal places: 4.500, 4.050, 4.505, 4.055.
All have 4 ones, so compare the tenths: 5, 0, 5, 0 – so 4.500 and 4.505 are bigger.
Compare these two: 4.505 > 4.500 (thousandths 5 > 0).
Answer : C) 4.505

10. 5.3 + 2.6 = **[A]**

A) 7.9 B) 7.8 C) 8.9 D) 7.19

[Chapter 3 : A Peek Beyond the Point]

Solution :

Add the tenths: 3 + 6 = 9 tenths.
Add the ones: 5 + 2 = 7.
5.3 + 2.6 = 7.9.
Answer : A) 7.9

11. Which of the following represents "4 less than a number x"? **[B]**

A) 4 − x B) x − 4 C) 4x D) x + 4

[Chapter 4 : Expressions using Letter-Numbers]

Solution :

"4 less than x" means 4 is taken away from x.
So the expression is x − 4 (not 4 − x).
Answer : B) x − 4

12. The simplified form of 3a + 5a − 2a is **[A]**

A) 6a B) 10a C) 8a D) 6

[Chapter 4 : Expressions using Letter-Numbers]

Solution :

All the terms are like terms (they contain a), so add and subtract the numbers.
3 + 5 − 2 = 6.
3a + 5a − 2a = 6a.
Answer : A) 6a

13. The perimeter of a regular pentagon of side s units is **[B]**

A) s + 5 B) 5s C) s^5 D) 5 − s

[Chapter 4 : Expressions using Letter-Numbers]

Solution :

A regular pentagon has 5 equal sides, each of length s.
Perimeter = s + s + s + s + s = 5s.
Answer : B) 5s

14. Two lines intersect at a point. If one of the four angles formed is 70°, the angle vertically opposite to it is **[B]**

A) 110° B) 70° C) 20° D) 290°

[Chapter 5 : Parallel and Intersecting Lines]

Solution :

Vertically opposite angles formed by two intersecting lines are equal.
So the angle vertically opposite to 70° is also 70°.
Answer : B) 70°

15. When two parallel lines are cut by a transversal, each pair of corresponding angles is [**A**]

A) equal B) complementary C) a linear pair D) unequal

[Chapter 5 : Parallel and Intersecting Lines]

Solution :

If two parallel lines are cut by a transversal, the corresponding angles are equal.
(Conversely, if corresponding angles are equal the lines are parallel.)
Answer : A) equal

16. Two angles form a linear pair. If one of them is 45°, the other is [**B**]

A) 45° B) 135° C) 55° D) 90°

[Chapter 5 : Parallel and Intersecting Lines]

Solution :

The angles of a linear pair add up to 180°.
Other angle = 180° − 45° = 135°.
Answer : B) 135°

17. The sum of two odd numbers is always [**B**]

A) odd B) even C) prime D) a multiple of 3

[Chapter 6 : Number Play]

Solution :

Try examples: 3 + 5 = 8, 7 + 9 = 16, 1 + 11 = 12 – all are even.
Reason: two odd numbers can be paired as complete pairs of objects, so nothing is left over; the sum is even (an odd + an odd = even).
Answer : B) even

18. The magic sum of a 3 × 3 magic square made using the numbers 1 to 9 is [**B**]

A) 12 B) 15 C) 18 D) 45

[Chapter 6 : Number Play]

Solution :

Sum of the numbers 1 to 9 = 1 + 2 + ... + 9 = 45.
The 9 numbers fill 3 rows, and every row has the same sum.
Magic sum = 45 ÷ 3 = 15.
Answer : B) 15

19. The sum of the three angles of a triangle is [**B**]

A) 90° B) 180° C) 270° D) 360°

[Chapter 7 : A Tale of Three Intersecting Lines]

Solution :

By the angle sum property, the three angles of any triangle add up to 180°.
Answer : B) 180°

20. Which of the following sets of lengths (in cm) can be the sides of a triangle? [**C**]

A) 3, 6, 9 B) 5, 10, 20 C) 10, 20, 25 D) 10, 20, 35

[Chapter 7 : A Tale of Three Intersecting Lines]

Solution :

Three lengths form a triangle only if the sum of any two is greater than the third (check the two smaller ones against the largest).
A) 3 + 6 = 9, which is not greater than 9 ✗.
B) 5 + 10 = 15, which is less than 20 ✗.
C) 10 + 20 = 30 > 25 ✓ (and the other two sums are bigger too).
D) 10 + 20 = 30, which is less than 35 ✗.
Answer : C) 10, 20, 25

Section B : Answer the following questions in your answer booklet. 4 x 2 = 8

21. Find 98 × 15 using the distributive property.

[Chapter 2 : Arithmetic Expressions]

Solution :

Write 98 as 100 − 2: 98 × 15 = (100 − 2) × 15. **[1 M]**

Distribute: (100 − 2) × 15 = 100 × 15 − 2 × 15 = 1500 − 30 = 1470. **[1 M]**

Answer : 98 × 15 = 1470

Total : 2 Marks

22. Remove the brackets and write the expression having the same value: (i) 25 − (8 + 6) (ii) 25 − (8 − 6)

[Chapter 2 : Arithmetic Expressions]

Solution :

(i) The bracket has a minus sign before it, so the signs of both terms inside change: 25 − (8 + 6) = 25 − 8 − 6. **[1 M]**

(ii) Here the signs change inside the bracket: 25 − (8 − 6) = 25 − 8 + 6. **[1 M]**

Check: (i) 25 − 14 = 11 = 25 − 8 − 6. (ii) 25 − 2 = 23 = 25 − 8 + 6.

Answer : (i) 25 − 8 − 6 (ii) 25 − 8 + 6

Total : 2 Marks

23. Fill in the blanks with "odd" or "even": (i) The sum of an odd number of even numbers is _____. (ii) The sum of an even number of odd numbers is _____.

[Chapter 6 : Number Play]

Solution :

(i) The sum of any number of even numbers is even. Example: 2 + 4 + 6 = 12 (three even numbers, sum even). **[1 M]**

(ii) Odd numbers can be paired up (an even number of them) and each pair adds to an even number. Example: 3 + 5 + 7 + 9 = 24 (even). **[1 M]**

Answer : (i) even (ii) even

Total : 2 Marks

24. Check whether a triangle can be formed with the following side lengths: (i) 4 cm, 5 cm, 8 cm (ii) 3 cm, 5 cm, 9 cm

[Chapter 7 : A Tale of Three Intersecting Lines]

Solution :

A triangle is possible only if the sum of the two smaller sides is greater than the largest side.

(i) 4 + 5 = 9, and 9 > 8, so a triangle can be formed. **[1 M]**

(ii) 3 + 5 = 8, and 8 is less than 9, so a triangle cannot be formed. **[1 M]**

Answer : (i) possible (ii) not possible

Total : 2 Marks

Section C : Answer the following questions in your answer booklet. 5 x 4 = 20

25. Read the following numbers in Indian place value notation and write their number names in both the Indian and American systems: (a) 4050678 (b) 48121620

[Chapter 1 : Large Numbers Around Us]

Solution :

(a) 4050678 – Indian notation: 40,50,678. **[½ M]**

Indian name: forty lakh fifty thousand six hundred seventy-eight. **[½ M]**

American notation: 4,050,678. **[½ M]**

American name: four million fifty thousand six hundred seventy-eight. **[½ M]**

(b) 48121620 – Indian notation: 4,81,21,620. **[½ M]**

Indian name: four crore eighty-one lakh twenty-one thousand six hundred twenty. **[½ M]**

American notation: 48,121,620. **[½ M]**

American name: forty-eight million one hundred twenty-one thousand six hundred twenty. **[½ M]**

Answer : (a) 40,50,678 – forty lakh fifty thousand six hundred seventy-eight; 4,050,678 – four million fifty thousand six hundred seventy-eight

Answer : (b) 4,81,21,620 – four crore eighty-one lakh twenty-one thousand six hundred twenty; 48,121,620 – forty-eight million one hundred twenty-one thousand six hundred twenty

Total : 4 Marks

26. Hasantika says that the sum of 4,63,128 and 4,19,682 is near 8,00,000, and Srestha says that it is near 9,00,000. Find the exact sum. Whose estimate is closer to the sum?

[Chapter 1 : Large Numbers Around Us]

Solution :

Exact sum: 4,63,128 + 4,19,682 (add column by column from the right). **[1 M]**

Units 8 + 2 = 10, tens 2 + 8 + 1 = 11, hundreds 1 + 6 + 1 = 8, thousands 3 + 9 = 12, ten-thousands 6 + 1 + 1 = 8, lakhs 4 + 4 = 8. Sum = 8,82,810. **[1 M]**

Difference from Hasantika's estimate: 8,82,810 − 8,00,000 = 82,810. **[½ M]**

Difference from Srestha's estimate: 9,00,000 − 8,82,810 = 17,190. **[½ M]**

17,190 is less than 82,810, so Srestha's estimate is closer. **[1 M]**

Answer : Exact sum = 8,82,810; Srestha's estimate (9,00,000) is closer.

Total : 4 Marks

27. Find the values of the following expressions and say whether the two expressions in each pair have the same value: (i) (6 + 10) − 2 and 6 + (10 − 2) (ii) 16 − (8 − 3) and (16 − 8) − 3

[Chapter 2 : Arithmetic Expressions]

Solution :

(i) (6 + 10) − 2 = 16 − 2 = 14 and 6 + (10 − 2) = 6 + 8 = 14. **[1 M]**

Both are equal, so the two expressions have the same value. **[1 M]**

(ii) 16 − (8 − 3) = 16 − 5 = 11 and (16 − 8) − 3 = 8 − 3 = 5. **[1 M]**

11 ≠ 5, so the two expressions do not have the same value. **[1 M]**

Answer : (i) 14 and 14 – same value (ii) 11 and 5 – different values

Total : 4 Marks

28. Create a 3 × 3 magic square using each of the numbers 2 to 10 exactly once, and write its magic sum. (Use the grid given below.)

[Chapter 6 : Number Play]

Solution :

Sum of the nine numbers = 2 + 3 + 4 + ... + 10 = 54. The three rows have equal sums, so the magic sum = 54 ÷ 3 = 18. **[1 M]**

The middle number of the square is one third of the magic sum: 18 ÷ 3 = 6 (the middle number of 2 to 10 is 6). **[1 M]**

Fill the grid (every row, column and diagonal must add to 18): **[1 M]**

9	**2**	**7**
4	**6**	**8**
5	**10**	**3**

Check: rows 9+2+7 = 18, 4+6+8 = 18, 5+10+3 = 18; columns 9+4+5 = 18, 2+6+10 = 18, 7+8+3 = 18; diagonals 9+6+3 = 18, 7+6+5 = 18. **[1 M]**

Answer : Magic sum = 18 (the magic square is shown above; any correct arrangement is accepted)

Total : 4 Marks

29. Two consecutive numbers in the Virahanka sequence are 987 and 1597. What are the next 2 numbers in the sequence? What are the previous 2 numbers in the sequence? (In the Virahanka sequence 1, 2, 3, 5, 8, 13, …, every number is the sum of the two numbers before it.)

[Chapter 6 : Number Play]

Solution :

Next numbers: 987 + 1597 = 2584. **[1 M]**

1597 + 2584 = 4181. **[1 M]**

Previous numbers: the number before 987 is 1597 − 987 = 610. **[1 M]**

The number before 610 is 987 − 610 = 377. **[1 M]**

Check: 377 + 610 = 987 ✓. The sequence is ..., 377, 610, 987, 1597, 2584, 4181, ...

Answer : Next two numbers: 2584 and 4181; previous two numbers: 377 and 610

Total : 4 Marks

Section D : Answer the following questions in your answer booklet. An internal choice is provided for each question. 4 x 8 = 32

30. A) (i) Find the sums: (a) 2.15 + 5.26 (b) 6.236 + 0.487 (2 + 2)

(ii) Find the differences: (a) 17 − 16.198 (b) 34.505 − 18.1 (2 + 2)

(Or)

B) (i) Arrange the following decimal numbers in descending order: 2.567, 2.675, 2.768, 2.499, 2.698 (4)

(ii) Pratap purchases 0.25 kg of beans, 0.3 kg of carrots, 0.5 kg of potatoes, 0.2 kg of capsicums and 0.05 kg of ginger. Calculate the total weight of the items he bought. (4)

[Chapter 3 : A Peek Beyond the Point]

Solution :

Option A – 8 Marks

(i)(a) Write the numbers with the decimal points in a line: 2.15 + 5.26. **[½ M]**

Hundredths: 5 + 6 = 11 → write 1, carry 1. Tenths: 1 + 2 + 1 = 4. Ones: 2 + 5 = 7. **[1 M]**

2.15 + 5.26 = 7.41 **[½ M]**

(b) 6.236 + 0.487: thousandths 6 + 7 = 13 → write 3, carry 1. **[½ M]**
Hundredths 3 + 8 + 1 = 12 → write 2, carry 1. Tenths 2 + 4 + 1 = 7. Ones 6 + 0 = 6. **[1 M]**
6.236 + 0.487 = 6.723 **[½ M]**
(ii)(a) 17 − 16.198: write 17 as 17.000 so that all numbers have 3 decimal places. **[½ M]**
Thousandths 10 − 8 = 2, hundredths 9 − 9 = 0, tenths 9 − 1 = 8, ones 16 − 16 = 0 (after borrowing). **[1 M]**
17 − 16.198 = 0.802 (check: 16.198 + 0.802 = 17.000) **[½ M]**
(b) 34.505 − 18.1: write 18.1 as 18.100. Thousandths 5 − 0 = 5, hundredths 0 − 0 = 0, tenths 5 − 1 = 4. **[1 M]**
Ones 4 − 8: borrow 1 ten → 14 − 8 = 6; tens 2 − 1 = 1. So 34.505 − 18.1 = 16.405. **[1 M]**
Answer : (i)(a) 7.41 (b) 6.723 (ii)(a) 0.802 (b) 16.405

Total : 8 Marks

Option B – 8 Marks
(i) Compare the ones: all numbers have 2 ones. Compare the tenths: 2.567 → 5, 2.675 → 6, 2.768 → 7, 2.499 → 4, 2.698 → 6. **[1 M]**
Largest tenths digit: 7 → 2.768 is the greatest. Smallest tenths digit: 4 → 2.499 is the smallest. **[1 M]**
2.675 and 2.698 both have 6 tenths; compare hundredths: 7 and 9, so 2.698 > 2.675. **[1 M]**
Descending order: 2.768 > 2.698 > 2.675 > 2.567 > 2.499. **[1 M]**
(ii) Total weight = 0.25 + 0.3 + 0.5 + 0.2 + 0.05 (write all with two decimals: 0.25 + 0.30 + 0.50 + 0.20 + 0.05). **[1 M]**
0.25 + 0.30 = 0.55 **[1 M]**
0.55 + 0.50 = 1.05 and 1.05 + 0.20 = 1.25 **[1 M]**
1.25 + 0.05 = 1.30 kg **[1 M]**
Answer : (i) 2.768, 2.698, 2.675, 2.567, 2.499 (ii) 1.3 kg

Total : 8 Marks

31. A) (i) Simplify: 3(8a − 5b) − 8a − 4b − 16 (4)
(ii) Add the expressions: 4d − 7c + 9 and 8c − 11 + 9d (4)

(Or)

B) (i) Subtract 9a − 6b + 14 from 6a + 9b − 18. (4)
(ii) One plate of Jowar roti costs ₹30 and one plate of Pulao costs ₹20. If x plates of Jowar roti and y plates of Pulao are ordered in a day, write an algebraic expression for the total cost. Find the total cost when x = 3 and y = 2. (4)

[Chapter 4 : Expressions using Letter-Numbers]

Solution :

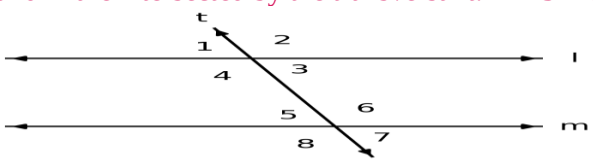
Option A – 8 Marks
(i) Remove the bracket: 3(8a − 5b) = 24a − 15b. **[1 M]**
The expression becomes 24a − 15b − 8a − 4b − 16. **[1 M]**
Group the like terms: (24a − 8a) + (−15b − 4b) − 16. **[1 M]**
= 16a − 19b − 16. **[1 M]**
(ii) Write the sum: (4d − 7c + 9) + (8c − 11 + 9d). **[1 M]**
Group the like terms: (4d + 9d) + (−7c + 8c) + (9 − 11). **[1 M]**
4d + 9d = 13d, −7c + 8c = c, 9 − 11 = −2. **[1 M]**
Sum = 13d + c − 2. **[1 M]**
Answer : (i) 16a − 19b − 16 (ii) 13d + c − 2

Total : 8 Marks

Option B – 8 Marks
(i) Subtraction means: (6a + 9b − 18) − (9a − 6b + 14). **[1 M]**
Remove the bracket by changing every sign inside it: 6a + 9b − 18 − 9a + 6b − 14. **[1 M]**
Group the like terms: (6a − 9a) + (9b + 6b) + (−18 − 14). **[1 M]**
= −3a + 15b − 32. **[1 M]**
(ii) Cost of x plates of Jowar roti = 30 × x = 30x. **[1 M]**
Cost of y plates of Pulao = 20 × y = 20y. **[1 M]**
Total cost = 30x + 20y (rupees). **[1 M]**
When x = 3 and y = 2: 30 × 3 + 20 × 2 = 90 + 40 = 130. **[1 M]**
Answer : (i) −3a + 15b − 32 (ii) 30x + 20y; total cost = ₹130

Total : 8 Marks

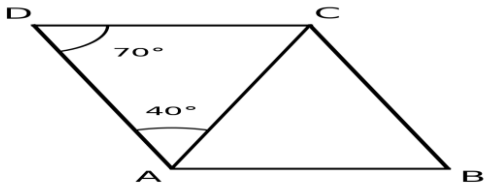
32. A) In the figure, the parallel lines l and m are intersected by the transversal t. If ∠3 = 65°, find the measures of:



(a) ∠1, ∠2 and ∠4 (3)
(b) ∠5, ∠6, ∠7 and ∠8 (4)
(c) Which angle is the alternate angle of ∠3? Is it equal to ∠3? (1)

(Or)

B) In the figure, AB ∥ CD and AD ∥ BC. If ∠DAC = 40° and ∠ADC = 70°, find ∠DAB, ∠CAB, ∠BCD and ∠ABC. (2 + 2 + 2 + 2)

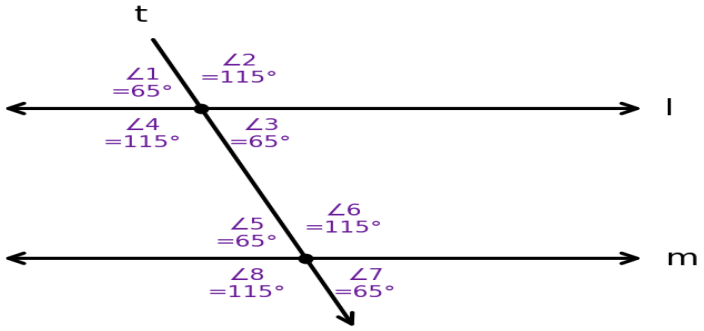


[Chapter 5 : Parallel and Intersecting Lines]

Solution :

Option A – 8 Marks

(a) ∠1 and ∠3 are vertically opposite angles, so ∠1 = ∠3 = 65°. **[1 M]**
∠2 and ∠3 form a linear pair, so ∠2 = 180° − 65° = 115°. **[1 M]**
∠4 and ∠2 are vertically opposite angles, so ∠4 = ∠2 = 115°. **[1 M]**
(b) ∠5 corresponds to ∠1 (l ∥ m), so ∠5 = 65°. **[1 M]**
∠6 corresponds to ∠2, so ∠6 = 115°. **[1 M]**
∠7 corresponds to ∠3, so ∠7 = 65°. **[1 M]**
∠8 corresponds to ∠4, so ∠8 = 115°. **[1 M]**



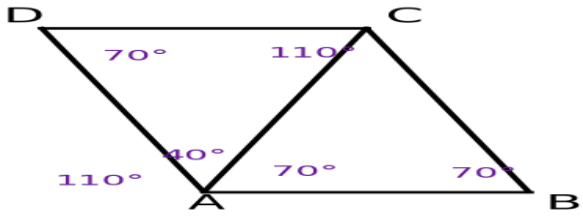
(c) The alternate (interior) angle of ∠3 is ∠5. Since l ∥ m, alternate angles are equal: ∠5 = 65° = ∠3. Yes, it is equal to ∠3. **[1 M]**

Answer : (a) ∠1 = 65°, ∠2 = 115°, ∠4 = 115° (b) ∠5 = 65°, ∠6 = 115°, ∠7 = 65°, ∠8 = 115° (c) ∠5; yes, equal

Total : 8 Marks

Option B – 8 Marks

Sum of the angles of triangle ADC: ∠DCA = 180° − ∠ADC − ∠DAC = 180° − 70° − 40° = 70°. **[1 M]**
AB ∥ DC and AC is a transversal, so the alternate angles are equal: ∠CAB = ∠DCA = 70°. **[1 M]**
∠DAB = ∠DAC + ∠CAB (C lies inside ∠DAB). **[1 M]**
∠DAB = 40° + 70° = 110°. **[1 M]**
AD ∥ BC and DC is a transversal, so the interior angles on the same side add up to 180°: ∠ADC + ∠BCD = 180°. **[1 M]**
∠BCD = 180° − 70° = 110°. **[1 M]**
AB ∥ DC and BC is a transversal, so ∠ABC + ∠BCD = 180°. **[1 M]**
∠ABC = 180° − 110° = 70°. **[1 M]**



Answer : ∠DAB = 110°, ∠CAB = 70°, ∠BCD = 110°, ∠ABC = 70°

Total : 8 Marks

33. A) (i) Construct a triangle ABC with BC = 5 cm, AB = 6 cm and CA = 5 cm. Write the steps of construction. (4)
(ii) In triangle PQR, ∠P = 50° and ∠Q = 65°. Find ∠R. If the side QR is extended to S, find the exterior angle ∠PRS. (4)

(Or)

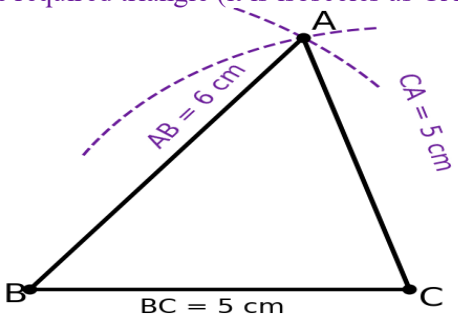
B) (i) Construct a triangle TRY with RY = 4 cm, TR = 7 cm and ∠R = 140°. Write the steps of construction. (4)
(ii) Which of the following sets of lengths can be the sides of a triangle? Give reasons. (a) 5 cm, 6 cm, 7 cm (b) 3 cm, 6 cm, 9 cm (c) 7 cm, 10 cm, 20 cm (d) 6 cm, 6 cm, 6 cm (4)

[Chapter 7 : A Tale of Three Intersecting Lines]

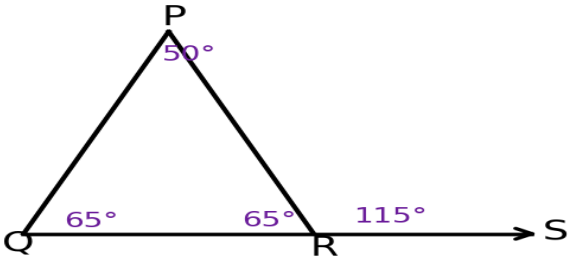
Solution :

Option A – 8 Marks

(i) Step 1: Draw a line segment BC = 5 cm. **[1 M]**
Step 2: With B as centre and radius 6 cm (= AB), draw an arc above BC. **[1 M]**
Step 3: With C as centre and radius 5 cm (= CA), draw another arc cutting the first arc. Mark the point of intersection as A. **[1 M]**
Step 4: Join AB and AC. ΔABC is the required triangle (it is isosceles as CA = CB = 5 cm). **[1 M]**



(ii) Angle sum property: ∠P + ∠Q + ∠R = 180°. **[1 M]**
∠R = 180° − 50° − 65° = 65°. **[1 M]**
Exterior angle property: the exterior angle ∠PRS = sum of the two interior opposite angles = ∠P + ∠Q. **[1 M]**
∠PRS = 50° + 65° = 115° (check: ∠PRS = 180° − ∠R = 180° − 65° = 115°). **[1 M]**

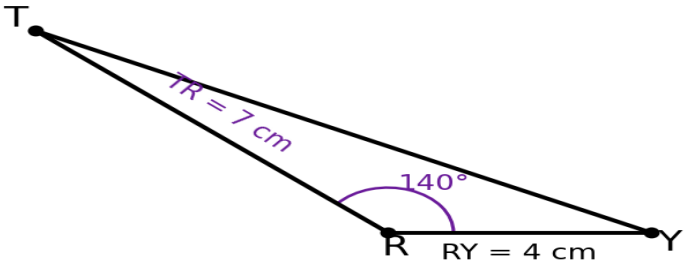


Answer : (ii) ∠R = 65° and exterior angle ∠PRS = 115°

Total : 8 Marks

Option B – 8 Marks

(i) Step 1: Draw a line segment RY = 4 cm. **[1 M]**
Step 2: At R, use a protractor to draw a ray RX making ∠YRX = 140°. **[1 M]**
Step 3: On the ray RX mark a point T such that RT = 7 cm. **[1 M]**
Step 4: Join TY. ΔTRY is the required triangle. **[1 M]**



(ii) The sum of any two sides must be greater than the third side.
(a) 5 + 6 = 11 > 7 (and 5 + 7 > 6, 6 + 7 > 5), so a triangle is possible. **[1 M]**
(b) 3 + 6 = 9, which is not greater than 9, so a triangle is not possible. **[1 M]**
(c) 7 + 10 = 17, which is less than 20, so a triangle is not possible. **[1 M]**
(d) 6 + 6 = 12 > 6 for every pair, so a triangle is possible (an equilateral triangle). **[1 M]**
Answer : (a) possible (b) not possible (c) not possible (d) possible

Total : 8 Marks

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